

# Was It Worth the Flip?

## A Monte Carlo Study of Stopping in Flip 7

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**Abstract**—Flip 7 is a card game of pure greed: keep flipping to score more, but draw a number you already hold and you lose the whole round. This study uses Monte Carlo simulation to find the best time to stop flipping in a one-player setting without action cards. For each stopping total from 1 to 63, 5000 rounds were simulated from a freshly shuffled deck, and the average points per round was recorded. The best average, 18.53 points per round, came from stopping at a total of 23, but the peak is flat: totals from 20 to 27 cannot be told apart from it. Beyond that range the risk of busting grows faster than the points gained, and by a total of 46 the average falls to about what stopping after one card earns. A simple rule of stopping in the twenties performs almost as well as the single best total.

### I. INTRODUCTION

The question I pose, while simple, touches on one very common behavior of humans, which is greed. And how better to showcase this than through one of my favorite card games, *Flip 7*. The goal of the game is to maximize your points, but draw the same numbered card from the deck, and you lose all your points that round, or *bust*. This can be seen as a game of luck, but it can also be seen as a counting game. But with so many cards to keep track of, and so many players, it gets hard to count. One can only look at one's cards and inherently choose whether or not another flip is worth it. The aim of this study is to find, using Monte Carlo simulation, the best time to stop flipping to maximize the average points per round in a one-player setting.

### II. GAME BACKGROUND

In each deck of *Flip 7* [1], there are the following:

- Number cards from 0 to 12, where each appears as many times as its number (e.g., 12 12 cards, 11 11 cards, and so forth). The exception is the 0 card, which appears once, but since it holds no value, it does not really matter.
- Action cards – Second Chance (if you bust, you are still able to continue the round), Freeze (forcing an opponent out of the round so they can't collect more points) & Flip 3 (forcing an opponent to flip 3 cards in a row, raising the likelihood that they might bust). **Action cards are not used in this simulation.**

A round ends in one of three ways. If the player flips a number already in their hand, they *bust* and score 0 for that round. If they stop, they bank the total of their hand. If they hold seven different numbers, the round ends with a bonus of 15 points.

### III. METHOD

This simulation focuses only on a one-player environment to find the best stopping total, defined as  $T$ , ranging from 1 to 63. The upper limit is the largest hand total possible with seven different numbers ( $12 + 11 + 10 + 9 + 8 + 7 + 6 = 63$ ), since the round ends at the seventh card. Every round starts from a freshly shuffled full deck, and the player follows one fixed rule: keep flipping until the hand total reaches  $T$ , then bank it.

A Monte Carlo simulation of  $n = 5000$  rounds was run for each  $T$ , and the scores were then averaged. In each round, the deck is shuffled, cards are flipped one at a time, and the round ends when the player busts, reaches seven different numbers, or reaches a total of at least  $T$ . The average score over the 5000 rounds estimates the expected points per round for that  $T$ .

Because each average comes from a finite number of rounds, it carries simulation noise. I measure this with the standard error,  $SE = s/\sqrt{n}$ , where  $s$  is the standard deviation of the round scores. The shaded band in the results figure shows the average  $\pm 2$  standard errors. A fixed random seed (47) makes the results reproducible.

### IV. RESULTS

Figure 1 shows the average points per round for each stopping total  $T$ . While  $T$  ranges from 1 to 63, the simulation suggests that the best stopping point is about a third of the way along that range, specifically at  $T = 23$ , where the average is 18.53 points per round across the 5000 simulated rounds. The shaded band in Figure 1 shows how much each average could shift if the simulation were rerun: where the bands of two stopping totals overlap, the simulation cannot tell their averages apart.

However, it should be noted that while  $T = 23$  is the best in this simulation, the peak of the curve is somewhat flat, and totals from 20–27 cannot be told apart from it before the average starts to dip. In a game, stopping when the total in hand is between 20 and 27 is therefore a reasonable choice, since the difference between these totals does not produce a meaningful change in the average score.

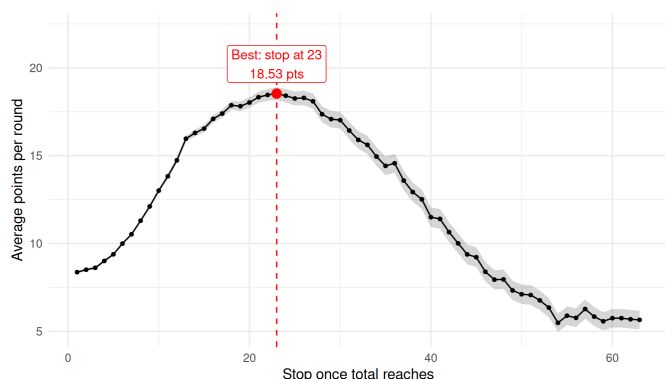


Fig. 1. Average points per round against stopping total  $T$ . The shaded band is the average  $\pm 2$  standard errors, and the red dot marks the highest average.

Beyond this range (20–27), the risk of busting grows faster than the points gained from one more card. For example, suppose the hand is worth 25 points. It likely contains at least one large card (e.g., 10, 11, or 12), and because each number appears as many times as its value, many copies of that card are still in the deck. Flipping again therefore carries a high chance of drawing a duplicate and losing the whole round.

In fact, from Figure 1, stopping at around  $T = 46$  yields about the same average score (8.39 points) as stopping after the first card (8.37 points), since the extra risk of busting cancels out the points gained.

## V. DISCUSSION

The main finding is that the best stopping total sits well below the maximum of 63, and that the peak is wide rather than sharp. At a low total, a flip is cheap: few numbers are in the hand, so few duplicates can hurt you. As the hand grows, two things happen at once. The hand is worth more, so a bust costs more, and it likely holds large numbers with many copies still in the deck, so a bust is more likely. Somewhere around 20 to 27 points, the expected gain from one more card stops outweighing the expected loss. By  $T = 46$ , the average has fallen to about what you would score by stopping after one card.

The flat peak matters in practice. Nobody at the table can compute  $T = 23$  exactly, but “stop somewhere in your twenties” is a rule a person can actually follow, and the simulation says it costs almost nothing compared to the single best total.

This study has clear limits:

- **Fixed total, not fixed situation.** A rule based only on the hand total ignores which numbers you hold and how many copies remain. Holding a 12 and a 3 is worth 15, but the risk of the next flip is very different from holding 7, 5 and 3. A smarter rule would stop based on the actual bust probability, and would likely beat any single  $T$ .
- **One player, no action cards.** Second Chance would push the best stopping point higher, because a bust is no longer final. Freeze and Flip 3 only matter against opponents.
- **Average points, not winning.** I maximised the expected score per round, but in a real game you do not want the

best average, you want to win. If you are far behind, a risky flip may be the right call even when the average says stop.

- **Simulation noise.** Each total was run for 5000 rounds with one fixed seed. The shaded bands show that nearby totals cannot be told apart, so  $T = 23$  should be read as the centre of a range, not a precise answer.

Future work could compare this fixed rule against one that stops based on bust probability, add the action cards back in, and test strategies against each other in multi-player games rather than against the deck alone.

## VI. CONCLUSION

In this simulation, the best time to stop flipping was at a total of 23, and anything from 20 to 27 did about as well. Beyond that, each extra card cost more in risk than it paid in points. The game rewards knowing when you have enough.

Yes, this is just a card game, and yes, it is a simple example. But how many times have we, as humans, wanted more when we already had enough? We often forget how privileged we are to be born in a place where we have access to food, education, and peace. And still, we want more. Like what? Building data centers that strain the environment we all depend on. Like the player holding 25 points who flips one more time, we risk what we already have for a little extra, except in real life, a bust is not as easy to reset as a new round.

## REFERENCES

- [1] The OP Games, “Flip 7.” [Online]. Available: <https://theop.games/pages/flip-7>. Accessed: Oct. 6, 2026.